# Supplementary Materials

## (a) Mean and relational effects of interaction terms

In the developmental system we model, when all interaction terms, $B$, are zero, changes to $G$ have an effect on the mean value of the phenotypic traits (including their sign) but not their correlation; hence we refer to $G$ as the 'direct' effects of the genotype on phenotypic characters. Conversely, for a distribution of $G$ elements each with a mean of zero, the effect of $B$ is to alter the correlation of phenotypic traits but not their mean. However, as expected for general epistatic models (and in contrast to statistical models that keep interaction terms and mean effects separate), whenever trait values are non-zero a change to an interaction term can have both a 'mean effect' as well as a 'relational effect'. This is important because interaction coefficients that have no mean effect on phenotype can only be selected because of their effect on the shape of the distribution of phenotypes they produce (Pavlicev, Cheverud & Wagner 2011), but interaction coefficients that also have a mean effect on phenotypes can be selected because of their mean effect but nonetheless have a (systematically related) effect on the shape of the distribution of phenotypes.

## (b) Hill-climbing model of selection

The evolution of the population is modelled by assuming the introduction of a mutant genotype, $G'$ and/or $B'$, being small mutations of $\bar{G}$ and $\bar{B}$, (in single copy number in a large population). This genotypic mutation uniquely determines a mutant phenotype, $P^{*\prime}$. In general, the probability of this mutant fixing in the population is proportional to the selective advantage of that mutation. But, here we aim to provide a phenomenological model of the evolution of development and as such it is the direction of selection, rather than quantified rates, that are important (rates will be sensitive to parameters such as magnitude of selection coefficients, mutation probabilities and the number of genetic loci contributing to a quantitative trait, etc. which will be case specific). Accordingly, it is sufficient for our purposes to assume a 'hill-climbing' model of selection (e.g., Kashtan et al. 2009), i.e., that the selection coefficient is sufficiently large that beneficial mutations fix and deleterious mutations do not. In general, mutation-rate-limited evolution can have qualitatively different dynamic properties to selection-intensity-limited evolution; but in all our experiments we find that the direction of selection using the former agrees with analysis using the latter (Pavlicev, Cheverud & Wagner (2011), and the stochastic nature of selection that would occur with small selection coefficients does not add to the clarity of exposition. Thus, if the mutant genotype is beneficial (i.e., *fitness*(*develop*($G'$, $B'$,$T$)) > *fitness*(*develop*($\bar{G}$,$\bar{B}$,$T$)), this becomes the mean genotype of the population ($\bar{G}(t+1)= G'$, $\bar{B}(t+1)=B'$ ), otherwise the loss of the mutant from the population leaves the mean genotype of the population unchanged. Only the fitness rankings of phenotypes (not the magnitudes of the fitnesses) are relevant for a hill-climbing selection model.

## (c) Effect of evolved interaction terms on the shape of phenotypic distributions

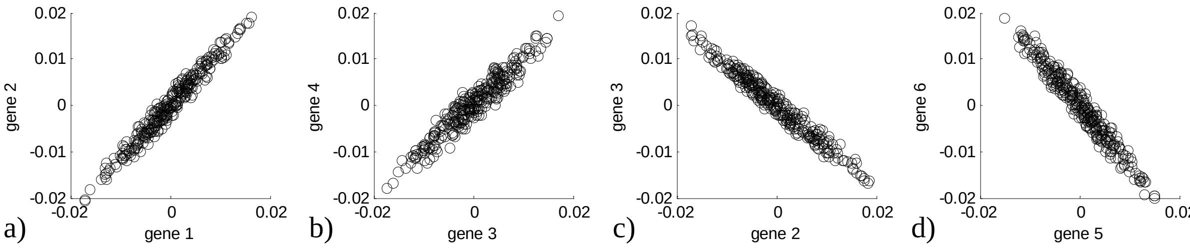


**Fig. S1**: The effect of evolved regulatory interactions (after 10,000 generations) on the correlation of expression (Experiment 1, target, $S$ = ++---+-+); for a pair of genes that are selected together $S_{12}$ = ++ (a), $S_{34}$ = -- (b), or selected contrariwise, $S_{23}$ = +- (c), $S_{56}$ = -+ (d). Each point in each distribution shows the expression levels of two genes from a phenotype developed using the evolved regulatory interactions. Each phenotype is developed from a random $G$ drawn from a uniform distribution.

Figure S1 confirms that the correlation of characters in the phenotypic distribution produced by the evolved developmental interactions agrees with the direction of selection experienced on those two characters in the selective environment.

### (d) The cumulative effect of many small mutations under directional selection.

Under *directional* selection, the cumulative effect of a large number of small mutations is equivalent to the effect of a small number of large mutations when controlling for variance. When the time for *G* to stabilise after a change in environment is small compared to the number of generations after it stabilises before the next change in environment, we can therefore model the effect of natural selection in any one selective environment as follows. For each element of *B* (in random order without replacement) we test for both positive and negative directional selection by applying hill-climbing selection on two mutations, one drawn from a distribution with mean $q$, the other with mean $-q$, each with standard deviation $\sigma$. Under directional selection, at most one of these mutations will be retained by selection. Experiments 3 and 4 use $q$=0.02 and $q$=0.005, respectively. In both cases we use a standard deviation $\sigma$=0.01$q$.

A standard deviation of 1% of the mean simulates the cumulative effect of $10^4$ discrete mutations (each with equal probability of being positive or negative) occurring before each change in environment (coming from $\sigma = \sqrt{np(1-p)},\ (p = 0.5)$). Naturally, there is a trade-off in these experiments between how many mutations occur before each change in environment and how many changes in environment are simulated in total. If the latter is increased, the former can be decreased, but this requires simulating more mutation-selection cycles. Here we find that 40 changes in environment is sufficient for Experiment 3 whereas Experiment 4 uses lower mutation magnitudes and 800 changes in environment (on average, 100 presentations of each training pattern). This low mutation rate enables the evolution of the 16 phenotypic attractors without 'over-fitting' to the 8 target patterns (see section (f)).

### (e) Developmental switching – abrupt change in phenotype from linear change in genotype

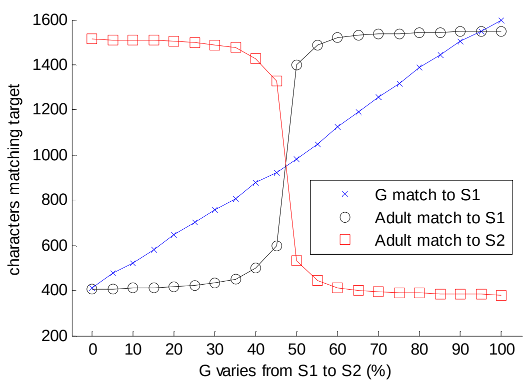


**Fig. S2**: Quantified detail of Fig.3.f. – match of *G* and adult phenotypes to targets *S1* and *S2*.

In an experiment where *G* varies systematically from *S1* to *S2* in steps of 5% (Fig. 3.f), note that as *G* changes slowly from *S1* to *S2* the adult phenotype switches abruptly from *S1* to *S2*.

### (f) Generalisation and evolvability

Any kind of canalisation seems to oppose the possibility that a developmental representation can be primed to produce phenotypic patterns that are genuinely new. However evolved correlations – or more exactly, the restrained application of evolved correlations – can facilitate evolutionary novelty in a quantifiable sense. Specifically, we can use knowledge of neural network learning to understand the affordances and limitations of enhanced evolvability to produce outputs that are new. Here we equate the selective environment that the evolving GRN has already been exposed to with the *training set* of the learning neural network. High fitness in these selective environments corresponds to high performance on the training set. Whereas, the ability to evolve high fitness phenotypes in *new* selective environments is analogous to the ability of a well-trained neural network to *generalise* to an unseen *test set*. For example, in Experiment 3, the 'training set', contains 8 patterns but we observe that the network is predisposed to create 16 phenotypic patterns, not just these 8.

This type of generalisation is not to be taken for granted, however, even when the training set contains the structural patterns that are representative of the general class. The problem is that the training set, being only a subset of the general class, necessarily also contains other structural patterns that you do not want the system to learn if it is to provide good generalisation. A learner may in some cases learn the specific patterns of the training set and thereby fail to generalise to the test set. This is called *over-fitting* (Rumelhart et al. 1986). In general, the difficult part of machine learning is not the task of getting high performance on the training set but the task of getting good generalisation on the test set. The issue is exacerbated if the model space is particularly 'expressive' or high-dimensional (as is necessary for complex learning tasks). In this case the model space will be highly *under-constrained* – meaning that there are many ways to represent the training set perfectly. Unfortunately, most of these solutions do not generalise well; so the task of finding general solutions is all the more difficult when the model space is high-dimensional.

Note that any kind of generalisation is equivalent to a 'failure' of a memory to represent *only* the training set. One solution to over-fitting is therefore to limit the complexity of the model space such that the learning system can only attain high fitness on the training set by finding general solutions; in other words, such that it does not have the capacity to learn the patterns by rote. In neural network models, model complexity is sometimes controlled by adding a penalty for each neural connection that is used (O'Reilly & Munakata 2000). Similarly, Clune, Mouret & Lipson (2013) add a cost to connections and observed improved evolvability, and Kashtan, Noor & Alon (2007) simply limit the number of logic gates a solution can use. In the current work, the training set of Experiment 3 has no (pairwise) correlations between modules and thus there is nothing for the GRN to learn. Or more exactly, there is no structure that the GRN is *capable* of learning. [2]

In the context of neural learning, when a memory is formed that was not in the training set this is sometimes referred to as a "spurious memory". But spurious memories can also be interpreted as a simple form of generalisation – producing a pattern that contains features of several different training patterns (Watson, Buckley & Mills 2009, Watson, Mills & Buckley 2010, Jang, Kim & Lee 1992). In some cases, this generalisation ability has the effect of identifying meso-scale features that are common to many patterns and putting them together in new combinations. This can create a genuinely new pattern that is substantially different from any of the training patterns, and yet contains features that resemble sub-patterns observed in multiple training patterns (Watson, Mills & Buckley 2010), as illustrated in Experiment 3. It is no coincidence that this kind of generalisation involves modules. Consider two traits in different modules; If generalisation is to be permitted then the GRN must not enforce a correlation between these two traits. In fact, for *any* two traits in different modules, the GRN knows *nothing* about what combinations of traits are fit. Accordingly, the only way for the GRN to know *anything* about past selective environments (whilst still allowing some traits to vary independently) is for it to learn correlations in approximately disjoint subsets. Learning correlations within modules usefully restricts the phenotypes that are produced, but learning correlations between modules opposes generalisation.

Note that learning the correlations within modules improves performance on the training set, but the only advantage of *not* learning the correlations between modules, allowing modules to vary independently of one another, is for improving generalisation and performance on the test set. This suggests that the selective pressures on learning what things go together are quite different from the selective pressure to learn what things to keep separate ('parcelation' rather than 'integration'; Wagner & Altenberg, 1996). This perspective indicates that the challenge for understanding the evolution of evolvability is closely related to the difficulty of learning general models from data – i.e., overcoming the problem of over-fitted

[2] In Experiment 3, the number of loops in the training patterns represent *odd parity* – i.e., the training samples have 1 or 3 modules. Odd parity is a generalisation of XOR (logical Exclusive OR) to multiple inputs that famously cannot be solved by local pairwise interactions. Since the developmental network can represent pairwise correlations, but not higher-order correlations, it cannot learn this set of training patterns without also producing patterns with 0, 2 and 4 loops.

solutions that give high-performance on the training set but fail to generalise to unseen test cases. And more specifically, the capability of expressive of G-P mappings to represent complex structural patterns is balanced by their tendency to over-fit to the past and thus fail to generalise to future selective environments.

Nonetheless, the equivalence between developmental memory and associative learning demonstrated in the current paper indicates that the evolution of a G-P mapping can provide substantive adaptive advantages in quite general scenarios. For example, the capability of recurrent neural networks to find locally optimal solutions to constraint optimisation problems (Hopfield & Tank 1986) is well known and associative learning can improve this ability (Watson, Buckley & Mills 2010, Watson, Mills & Buckley 2011). Specifically, associative learning can improve the ability of a network to find high-quality solutions even when the problem has only 'accidental' structure rather than an explicit modular decomposition (Watson, Buckley & Mills 2011). But the advantage is much more pronounced when there is modular structure that can be exploited (especially if the G-P map changes the variational neighbourhood, enabling large but controlled variation in phenotype space) (Watson, Mills & Buckley 2011, Mills 2010, Mills & Watson 2009, Mills & Watson submitted). The potential for evolution by natural selection, given developmental processes with suitable heritable variation, to exhibit such adaptive capabilities suggests that the evolution of evolvability may alter the adaptive capabilities of natural selection in a substantial manner; and, perhaps more importantly, the application of a machine learning framework enables us to quantify and understand the affordances and limitations of this ability (Watson et al. 2010a).

### (g) Modularly varying environments produces modular structure in the interaction network

In Experiment 3 in the main text the two patterns in each module are arbitrary, not the complement of one another, and different patterns are used in each module (e.g., different rotations of loops). The characters that correspond to a block are also not continuous in the linear genome *G* (there is nothing about the experiment that is sensitive to this ordering of the phenotypic characters[3]). This serves to illustrate that the result does not require special symmetries. Here we provide an additional experiment, analogous to Experiment 3, but with more transparent modularity that can be easily verified by inspecting *B*. We studied 8 target phenotypes, each of which is composed of 16 characters, divided into four groups of four (Fig. S3.a). The four characters within each group only appear in the set of target patterns as either ---- or ++++. Four of these sub-patterns are combined to create a complete training pattern of 16 characters such that each target pattern contains one module of the "++++" type and the remainder are of the "----" type or vice versa.

Figure S3.b-c shows that the evolved correlations clearly identify the strong correlations amongst the genes within each module as predicted by Hebb's rule. We then examine the distribution of adult phenotypes produced from these evolved correlations (Fig. S3.d-e). We see that development reproduces only well-formed modules (i.e., the traits within each module match either '++++' or '----'), but that in addition to the target phenotypes, the adult phenotypes also exhibit other patterns produced by new combinations of those modules. In fact, all 16 possible combinations of the four modules are exhibited in the distribution of adult phenotypes (not just the 8 in the training set). In this simple example we can see the modularity in *B* (Fig. S3.b) (representing the fact that there are strong correlations between traits in the same module and no evolved correlations between traits in different modules) that thus allows new combinations of modules to be produced by development.

[3] The ability of development to produce novel phenotypes via new combinations of developmental modules is closely analogous to the ability of sexual recombination to produce novel genotypes via new combinations of genetic building blocks (Watson, Weinreich & Wakeley 2011), and the test case used here is structurally equivalent. However, the success of sexual recombination requires that genetic linkage corresponds with epistatic dependencies, whereas here useful modules can be identified and reproduced in new combinations regardless of the ordering of traits in the target phenotypes.

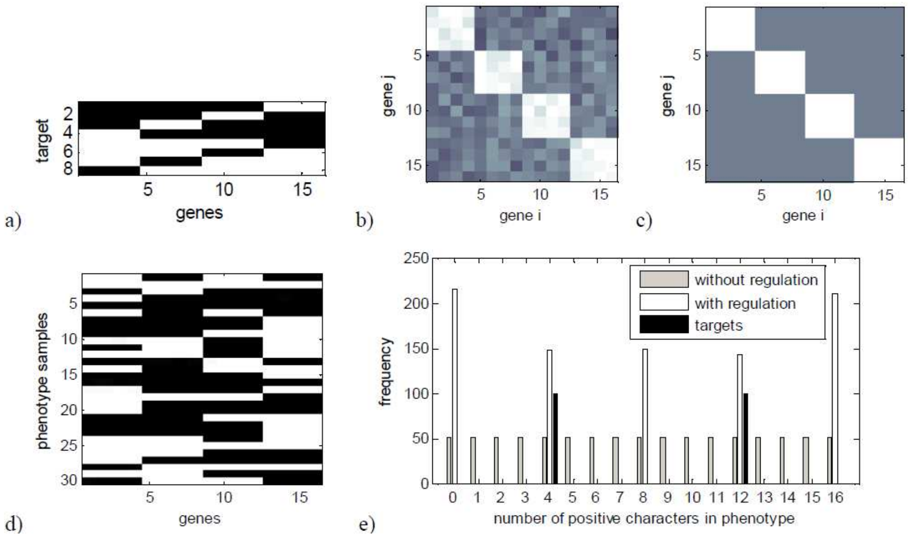


**Fig. S3:** Interaction coefficients evolved in modularly varying selective environment ($9.6 \cdot 10^6$ generations) and resultant phenotypes. a) The set of modularly varying target phenotypes. b) Evolved interaction coefficients. c) Hebbian interaction matrix summed over the 8 patterns. Evolved within-module interactions are positive and between-module interactions are approximately zero, exactly as expected under Hebbian learning. d) 30 example adult phenotypes developed from uniformly random *G*. Note that these include patterns that are not in the target set. e) Distributions of phenotype patterns; 'Targets' have either 4 or 12 positive traits. 'Without regulation' shows the frequency positive traits in *G*. For this figure only, *G* is are generated to be uniformly distributed in number of positive characters [0,16]; 'With regulation' shows the frequency '+' traits in the adult phenotype patterns developed with the evolved interaction coefficients. Note that the phenotypes produced include the target patterns (with one or three blocks of positive traits) but also all other combinations of modules (with zero, two and four blocks of positive traits).

### (h) Brief investigation of robustness to environmental noise

Each row is a sample of 10 phenotypes (as per Fig. 3.d) but each row shows the effect of environmental perturbation (in this extreme case, a full randomisation of the values of *P*) at progressively later time steps of development. For example, when environment affects *P* only 3 time steps from the end of development, the adult phenotype (at time step 10) is unable to recover either of the target phenotypes accurately.

However, note that since there are only two phenotypic attractors in this developmental system, and any *G* leads to either one or the other (Fig. 3.d), if development were allowed to continue for additional time steps after an environmental perturbation, in every case it will eventually equilibrate at one or other of the two targets as before. In these experiments we are interpreting environmental perturbation simplistically; a) we are investigating whether an adult phenotype can recover from an environmental perturbation not whether a phenotype can resist environmental perturbation in the first place, b) here we are also not investigating what happens when environmental perturbation of the phenotype is partial.

### (i) Robustness without reducing phenotypic variability

An evolved developmental process with appropriate flexibility may enable the recovery of high-fitness phenotypes quickly and reliably after a change in environment. The converse is also true, however; the ability of a population to evolve a particular phenotype may be retarded by a developmental process that does not produce appropriate phenotypic variation. This has some relevance to the apparent tension between evolvability and robustness. That is, the conservative and myopic quality of natural selection favours developmental canalisation and robustness which seems intrinsically opposed to an increase in phenotypic variability or evolvability. Some view these notions as two-sides of the same coin (Kirchner & Gerhart 1998, Draghi et al. 2010, Brigandt 2007); i.e., a predisposition to evolve some phenotypes more readily goes hand in hand with a decrease in the propensity to produce other phenotypes. Indeed, the regulatory connections that evolve in our model do not evolve because they increase the speed of future evolution, although they do, but rather because, in the immediate term, they increase the strength with which the current phenotype is expressed.

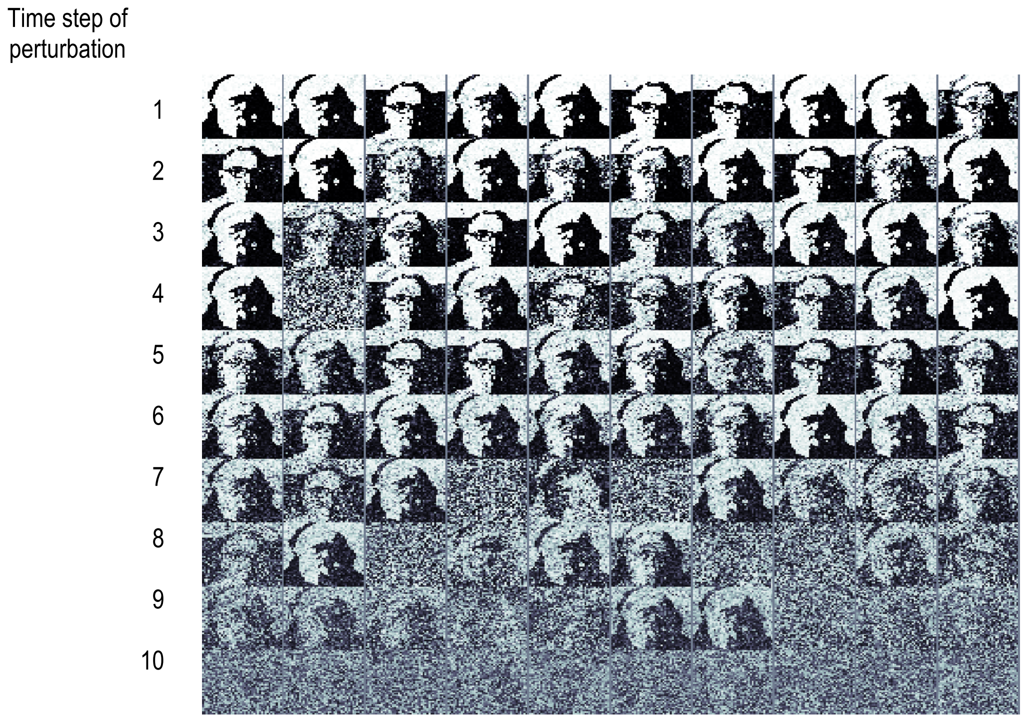


**Fig. S4:** Example phenotypes at developmental time step 10, after an environmental perturbation (randomisation) to the developing gene expression pattern at successively later developmental time steps (each image is an independent simulation, 10 examples for each perturbation period).

A multi-dimensional concept of variability, including correlated variability, provides a more sophisticated appreciation of the manner by which canalisation might oppose or facilitate evolvability. For example, if robustness is increased by canalisation applied uniformly in all dimensions then that would oppose evolvability. But there are (at least) two other possibilities. i) An increase in robustness might occur by reducing variability in some traits whilst leaving others free to vary; focussing the remaining variability on more useful dimensions. This partly alleviates the apparent tension of robustness and evolvability. ii) But more interestingly, evolved correlations, uniquely, can reduce the dimensionality of phenotype space by reducing the *combinations* of traits that occur in phenotypes, without reducing phenotypic variability in any *individual* phenotypic characters. For example, the two phenotypes ++ and -- constitute only half of the four possible phenotypes composed of two bi-allelic characters, but each individual trait can still take either value + or - (this is the developmental analogue of altering the linkage disequilibrium of simultaneously segregating alleles without altering allele frequencies/fixing alleles at either locus). Limiting character combinations without limiting the variability of the individual characters alleviates the tension between robustness and evolvability in a more profound sense.

The logic functions IFF and XOR correspond to positive and negative correlation, respectively. These functions are significant in machine learning because the output is not a linearly decomposable function of the inputs (McClelland & Rumelhart 1986). It is necessary to learn correlations to represent these functions, and representing these functions is necessary to produce a memory that can store more than one pattern without simply blending them. Simpler types of canalisation, e.g. from a linear developmental process, preclude this possibility (as illustrated in Figs. 3.h & 4.d).

**(j) Recall from partial stimuli showing developmental time**

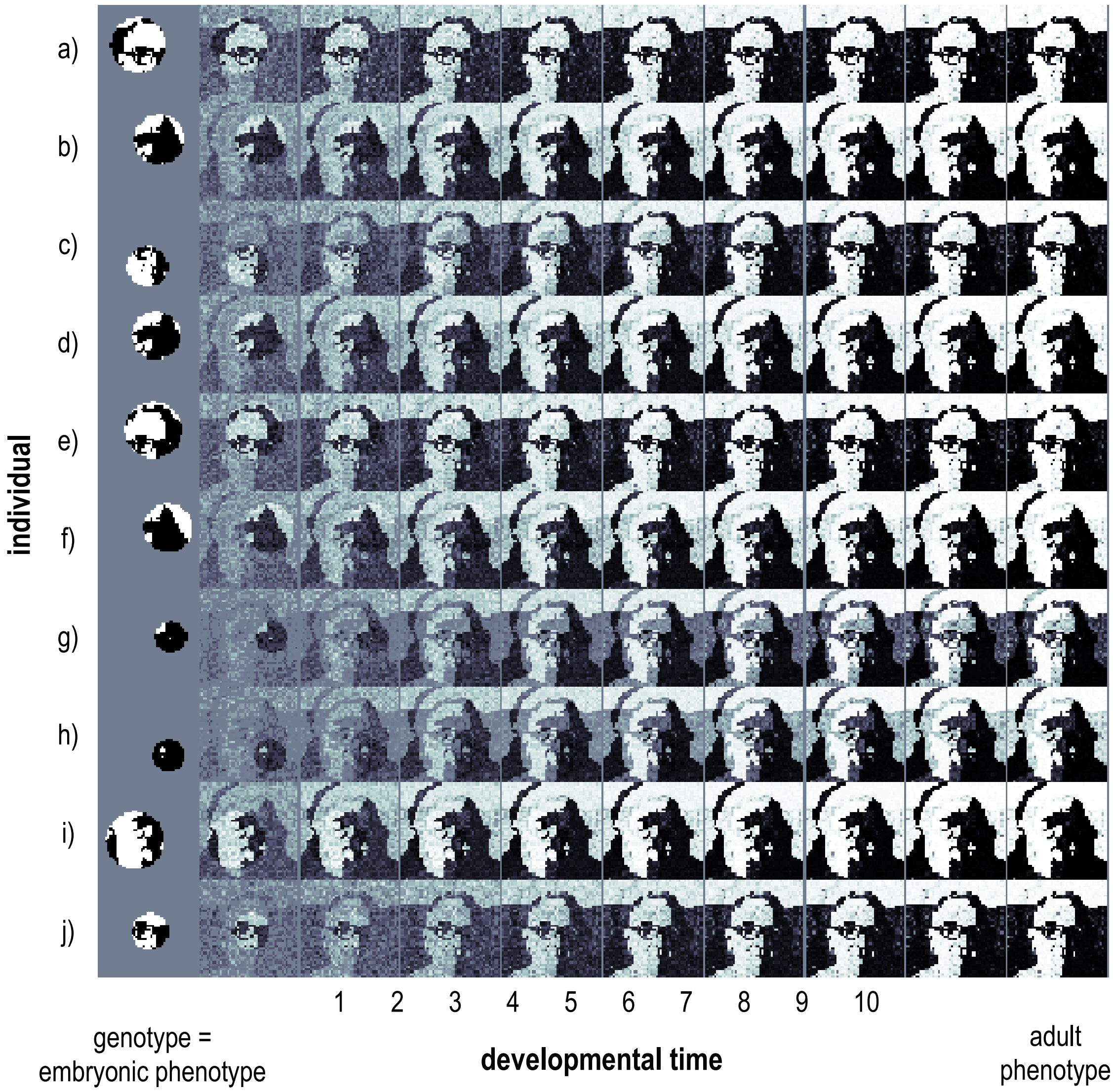


**Fig. S5: An evolved gene regulation network exhibits a developmental memory of phenotypes that have been selected for in the past.** The gene network was evolved in an environment where selection varies between two target phenotypes; an image of Charles Darwin and an image of Donald Hebb (the neuropsychologist from whom Hebbian learning takes its name). Each pixel in the image corresponds to the expression level of a gene. The evolved network of regulatory interactions introduces phenotypic correlations that create developmental attractors corresponding to these two phenotypes. We find that this is functionally equivalent to the manner in which a neural network learns an associative memory of a set of training patterns via Hebbian learning. This experiment shows that if the genotype, controlling the embryonic phenotype, partially resembles one of the targets, development 'recognises' which phenotype it belongs to and recreates the complete adult phenotype in a self-consistent manner. The development of ten example individuals from different genotypes (a-j) are shown over ten developmental time steps. In most cases development produces an adult phenotype that fully resembles only one of the phenotypes selected in the past. In cases where the genotype is almost equally matched to both phenotypes (e.g. g and h) the process of development breaks this symmetry, producing an adult phenotype that largely resembles only one, but some residual trace of the other phenotype is visible late into developmental time.

**See main text for references.**